The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction: The Unified 11-Part Search-and-Rescue Plot

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 Axiomatic Metaphysics and the Causal Tracking Loop

We formalize the system trajectory of an asset as a higher-order relational tracking loop:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{1}$$

where u represents environmental Navier-Stokes advection forces, m represents the external divine control input vector, y represents human or machine output behaviors, and E represents the absolute spiritual evaluation function.

2.1 Corollary 1.1: Anthropological Incompleteness and Vacuum Collapse

When the external control vector m is stripped from the network, human judgment operates as a strictly closed formal system, F. By Gödel's First Incompleteness Theorem, a closed system cannot generate its own semantic validation ($F \nvdash (v_{\text{human}} = v)$). Syntactic legalism remains blind to absolute truth, causing a critical valuation $v_k \in V'$ to become structurally unobtainable:

$$\exists v_k [V'_{v_k} \wedge \neg h(u, v_k)] \tag{2}$$

Without the active insertion of the divine control input vector, the spatiotemporal execution plane collapses into an empty relational set ($[\,]$)—a mathematical vacuum. Christ operates as the external Meta-Language that breaks this internal syntax boundary limitation.

3 Graph Neural Networks over the Ecclesial Social Manifold

The social network layer bridges the gap between W_2 and W_1 by mapping human interaction topologies to a Graph Neural Network (GNN). Nodes represent individual human agents and edges represent relational communication channels.

3.1 Mathematical Formulation of the Message-Passing Layer

The network updates node state representations recursively across distributed overlapping simplices (the body metaphor) using a Graph Convolutional Network (GCN) update function:

$$h_i^{l+1} = \text{ReLU}\left(W^l \frac{1}{\sqrt{\deg_i}\sqrt{\deg_j}} \sum_{j \in \mathcal{N}_i} h_j^l \right) \tag{3}$$

where W^l represents the trainable layer weight tensor matrix, and $\mathcal{N}_i$ represents the local neighborhood of node i.

3.2 The 27-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node at-

tribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 27}$. For each node i, the vector decomposes into three distinct socio-pragmatic tensor blocks:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{empirical}} \mid \boldsymbol{c}_i^{\text{cognitive}} \mid \boldsymbol{s}_i^{\text{institutional}} \right] \tag{4}$$

where $\boldsymbol{e}_i \in \mathbb{R}^{18}$ encapsulates the Barna loop splits, the 72% Pew United States Church macro-advection wavelets, and the JAX spatial grid coordinates. The cognitive component $\boldsymbol{c}_i \in \mathbb{R}^6$ packs the compiled LISP bar preferences and the 3-dimensional Rational Speech Act (RSA) scalar implicature distributions. The institutional component $\boldsymbol{s}_i \in \mathbb{R}^3$ isolates the zero-uncertainty Searlean speech-act flags (X counts as Y in C), allowing the network to collapse giant graph topologies down to a unified virtual 0-simplex token ready for decision tree lookup routing.

4 Three-Tier Nested S-Expression Grammars and Searlean Realities

We formalize the cross-compilation of continuous pathfinding metrics, recursive PPL choice distributions, and institutional status functions within a unified three-tier LISP S-Expression syntax grammar.

4.1 The Combined Grammar Tokenization Rules

The lexical State Machine tokenizes the input alphabet by mapping subatomic type hierarchies across three distinct conceptual domains:

1. **Grammar 1 (Kinematic Motion):** Evaluates continuous float vectors over non-convex keep-out zones: program ::= (go vector) | (goto object) | (if boolean action action).

2. **Grammar 2 (Probabilistic Social Reasoning):** Evaluates recursive social mentalizing via level-k sampling stencils: statement s ::= a : chooses($x \in \mathcal{X}$, wpp $= e$) | a : thinks[s^*].

3. **Grammar 3 (Searlean Institutional World):** Extends the execution environment to include illocutionary force operators: Assertives, Directives, Commissives, Expressives, and Declarations.

4.2 The Status Function Declaration Operator

A Speech-Act declaration explicitly bypasses raw sensory ambiguity (`observes`), invoking a collective ontological inception that instantiates a new institutional property assignment directly inside the decision forest leaves:

$$\mathcal{D}_{\text{inception}}\Big(\langle C,\, X,\, Y\rangle\Big) \implies (X \text{ counts as } Y \text{ in } C) \tag{5}$$

When a misplaced branch or subject error is triggered, the DSL agent recognizes the discrepancy and back-propagates structural look-up repairs, grafting a corrected zero-uncertainty exemplar node into the living document forest branch.